



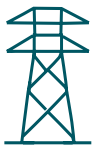
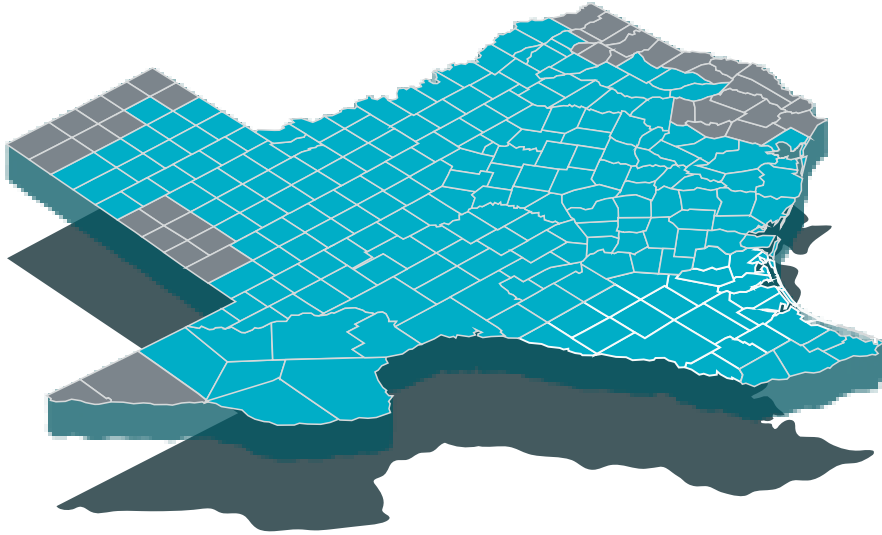
The Challenges of Ensuring Grid Reliability in the Era of Large Computational Loads

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ERCOT Facts



~\$14 billion Transmission projects endorsed in 2025



1,460+ generating units, including PUNs



1,928+ active Market Participants that generate, move, buy, sell or use wholesale electricity

ERCOT Records

Peak Demand Record

85,508 MW
(August 10, 2023)

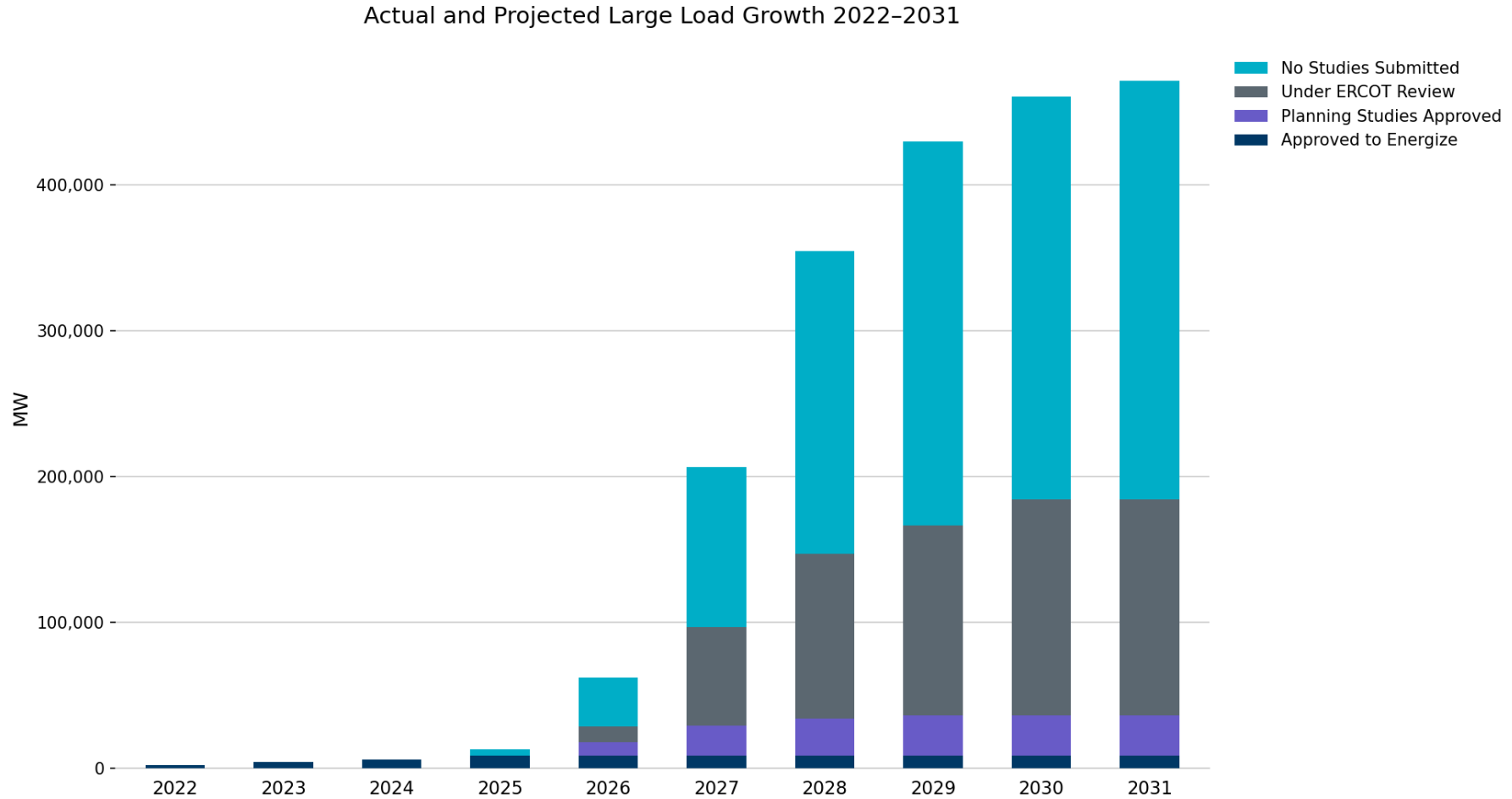
Weekend Peak Demand Record

85,116 MW
(August 20, 2023)

Expected Capacity for Summer 2026

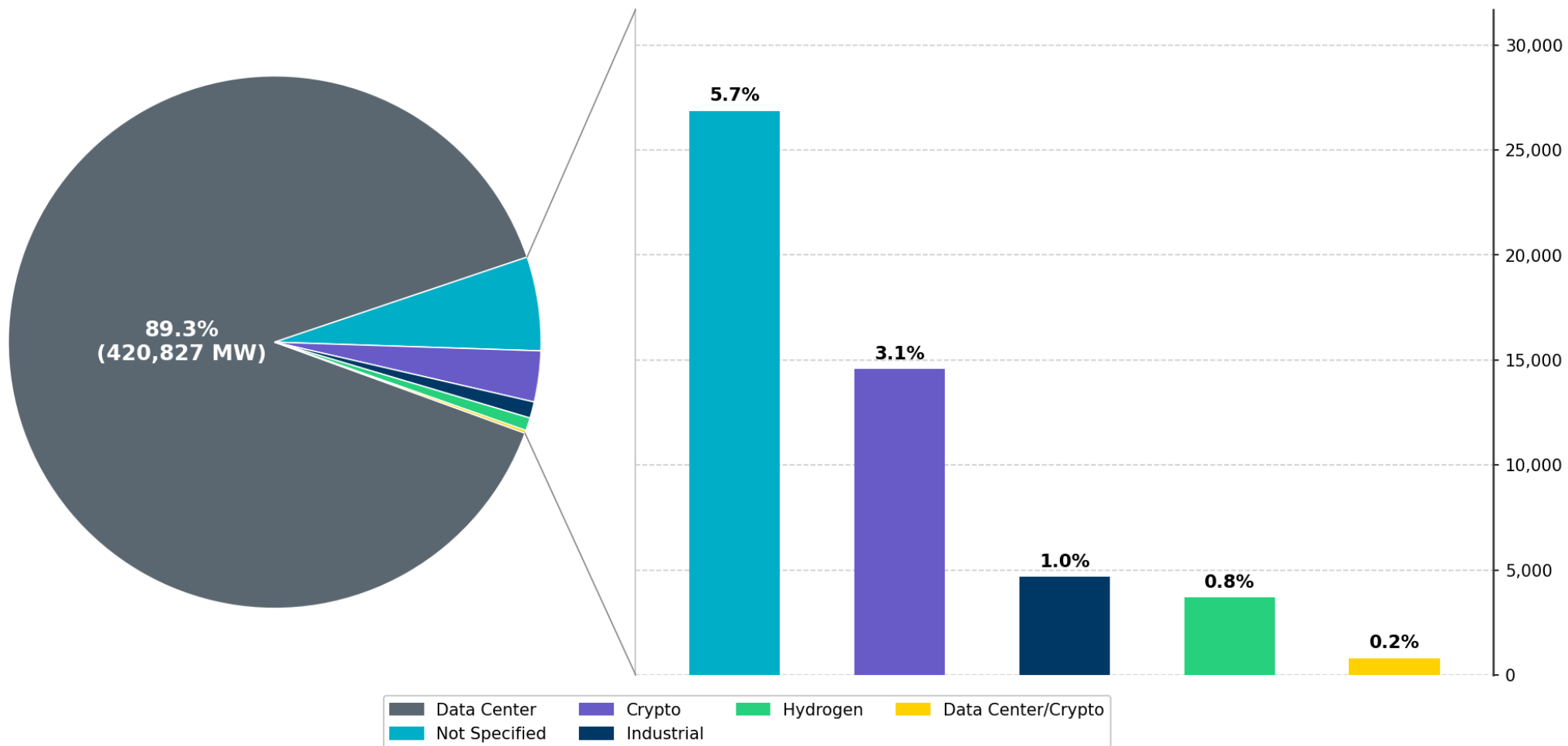
104,850+ MW
(December 2025 *Capacity, Demand and Reserves report*)

ERCOT Large Load Queue



**Numbers above only reflect LLs that have gone through ERCOT's Large Load Interconnection process

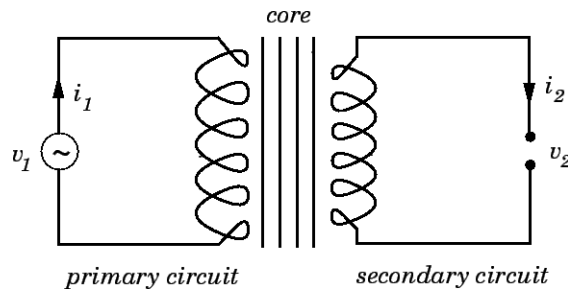
Large Load Project Distribution by Type



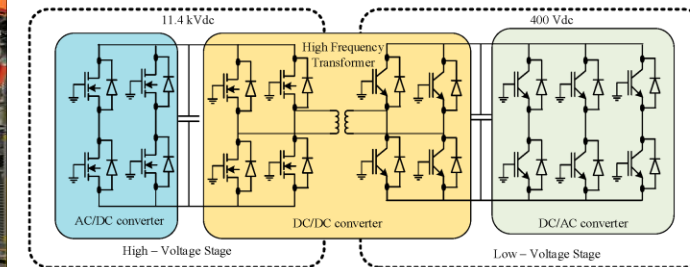
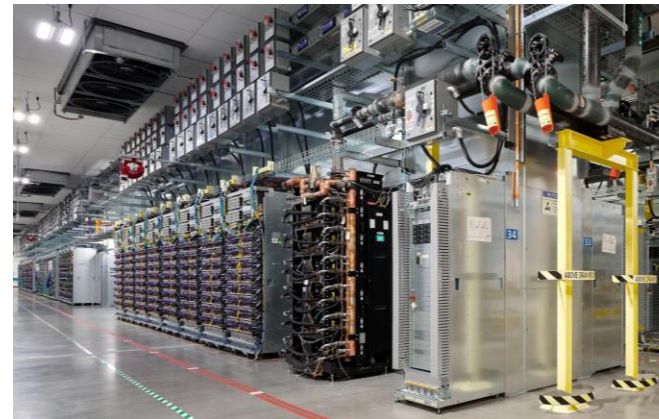
**Majority of not specified type expected to be data centers and crypto

The Changing Power System

Then – distributed load across residential, commercial, industrial users



The Future – power electronic based single-site loads rivaling regional demand



The Impact of Large Computational Loads

Across the world we're seeing increasing growth in Large Computational Loads (LCLs) such as data centers and cryptocurrency mining.

These facilities have several characteristics previously unseen on the bulk power system for most loads:

- Peak demands eclipsing any load previously connected to the system
- Power electronic interfaces with extremely fast response times
- Software controlled demand and performance characteristics

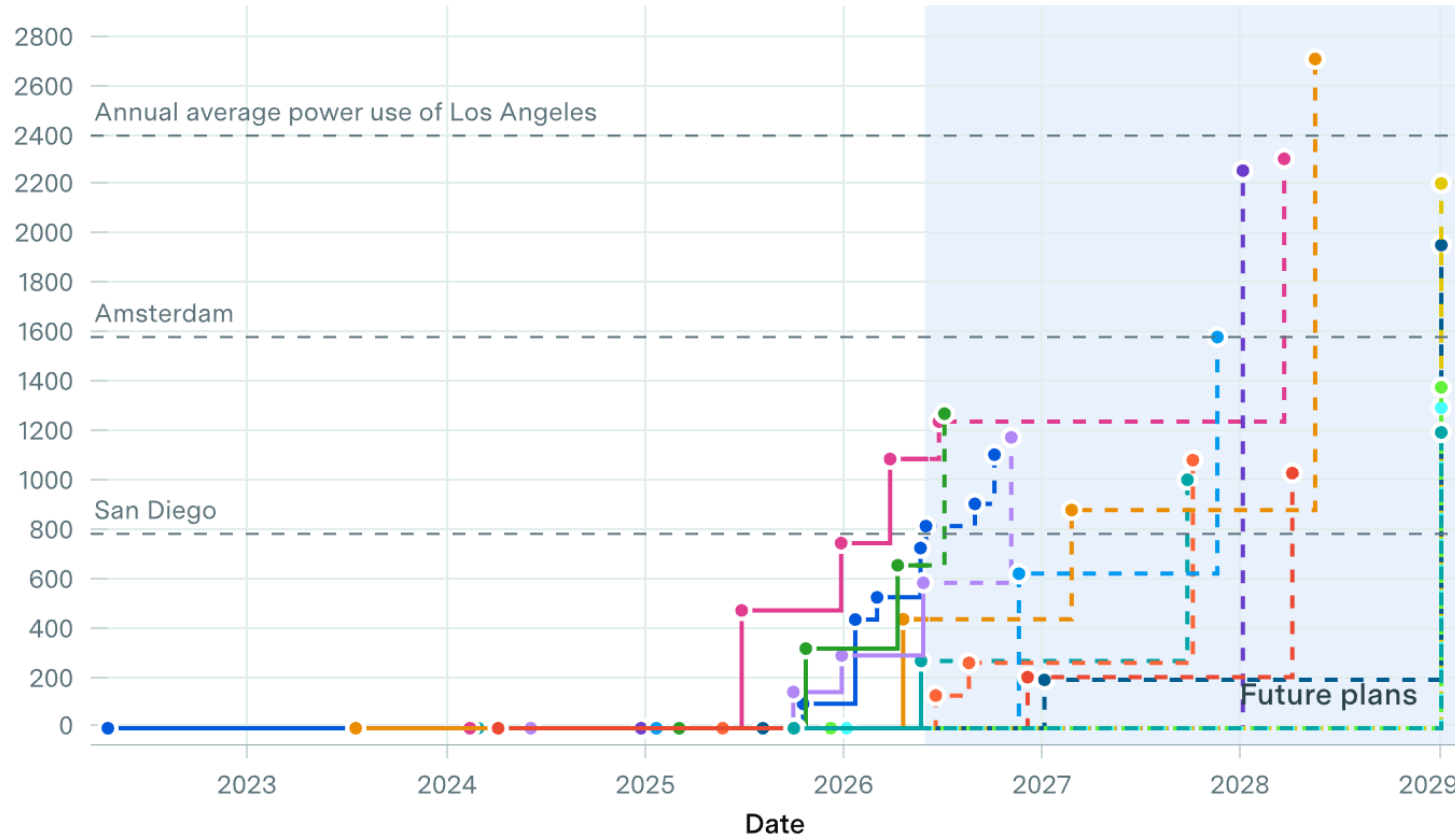
The disconnection of a data center can pose risks to grid reliability in the form of high frequency and voltages. This can potentially trigger overfrequency and overvoltage protection that could lead to a cascading outages across the system.

Additionally, repetitive patterns of load consumption from a data center can cause oscillations on the grid which can damage equipment such as generators.

Gigawatt Scale Data Centers

Frontier Data Centers

Power (MW)



Data centers (15/50 selected)

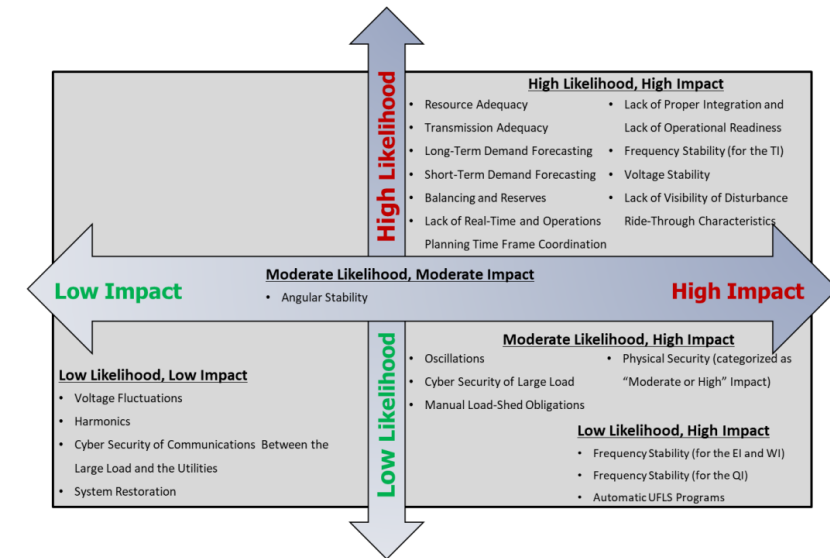
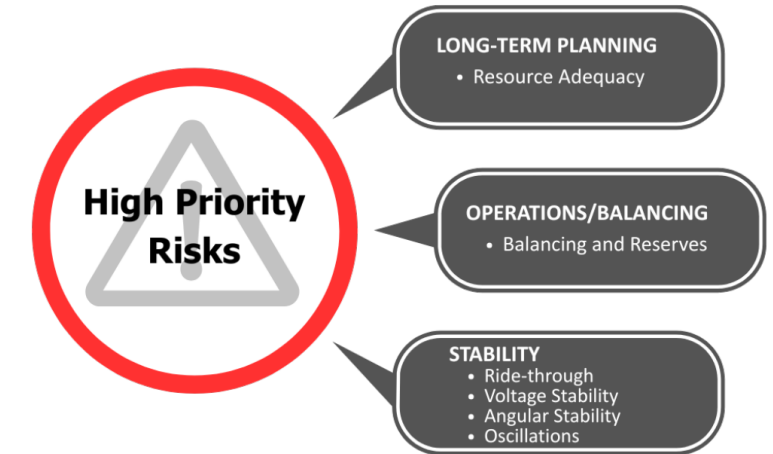
- Amazon Ridgeland
- Anthropic-Amazon New Carlisle
- Goodnight
- Meta Hyperion
- Meta Prometheus
- Microsoft Fairwater Wisconsin
- OpenAI Stargate Abilene
- Colossus 2
- QTS Cedar Rapids
- OpenAI Stargate Shackelford
- Google Fort Wayne
- OpenAI Stargate New Mexico
- OpenAI Stargate Wisconsin
- OpenAI Stargate Michigan
- OpenAI Stargate Milam

Reliability Challenges

Large Computational Loads pose several key challenges to maintaining Bulk Power System reliability.

- A disturbance resulting in load reduction or tripping can threaten grid stability.
- LCLs demand can vary rapidly causing oscillations that damage generation and posing forecasting challenges.
- Protection and grid control systems may not respond as expected due to inadequate load models and simulation results.
- Rapid LCL ramps stress reserves and the ability to balance the grid.
- LCLs can be built faster than generation and transmission can be deployed.

Key Takeaway: Of all these challenges ensuring uniform disturbance performance responses, preventing oscillations, and ensuring the ability to properly model the load in grid studies are among the most important challenges to address.



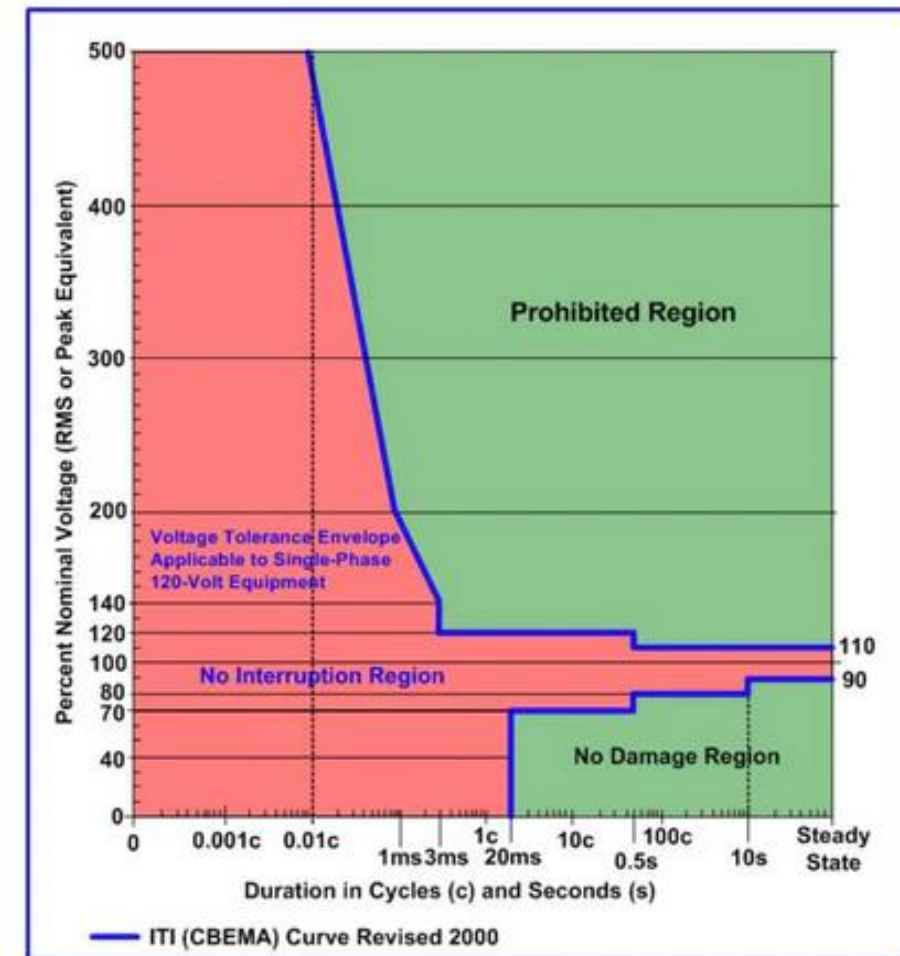
Disturbance Performance

The disconnection of a Large Load could cause high frequencies and voltages. This can potentially trigger overfrequency and overvoltage protection that could lead to cascading outages across the system.

In the context of LCLs, disturbance performance describes the behavior of the facility in response to grid disturbances. There are three critical aspects of disturbance performance for which a lack of requirements threatens grid stability.

- Voltage Ride-Through
- Frequency Ride-Through
- Post Fault Active Power Recovery

Today most electronics and data center equipment is designed to the ITIC curve. Preliminary analysis by grid operators across the world is showing that this behavior will not be sufficient enough to prevent a grid collapse on its own. After a fault we also need the LCL to return to pre-fault power consumption quickly or else frequency stability may be jeopardized.



Protection Setting Considerations

Another aspect of disturbance performance is protection system coordination. Some facilities are using protection schemes that count the number of voltage disturbances in a given timeframe. This logic needs to consider transmission breaker operations and normal reclosing practices.

Protection schemes must also be updated to not trigger on normal transmission high speed reclosing (HSR) or normal faults.

- HSR typically makes an attempt to reclose after the initial fault in 20-30 cycles range. Then 1-3 timed reclosing attempts follow HSR typically between 5 to 20 seconds. A normal fault typically clears in 3-6 cycles.

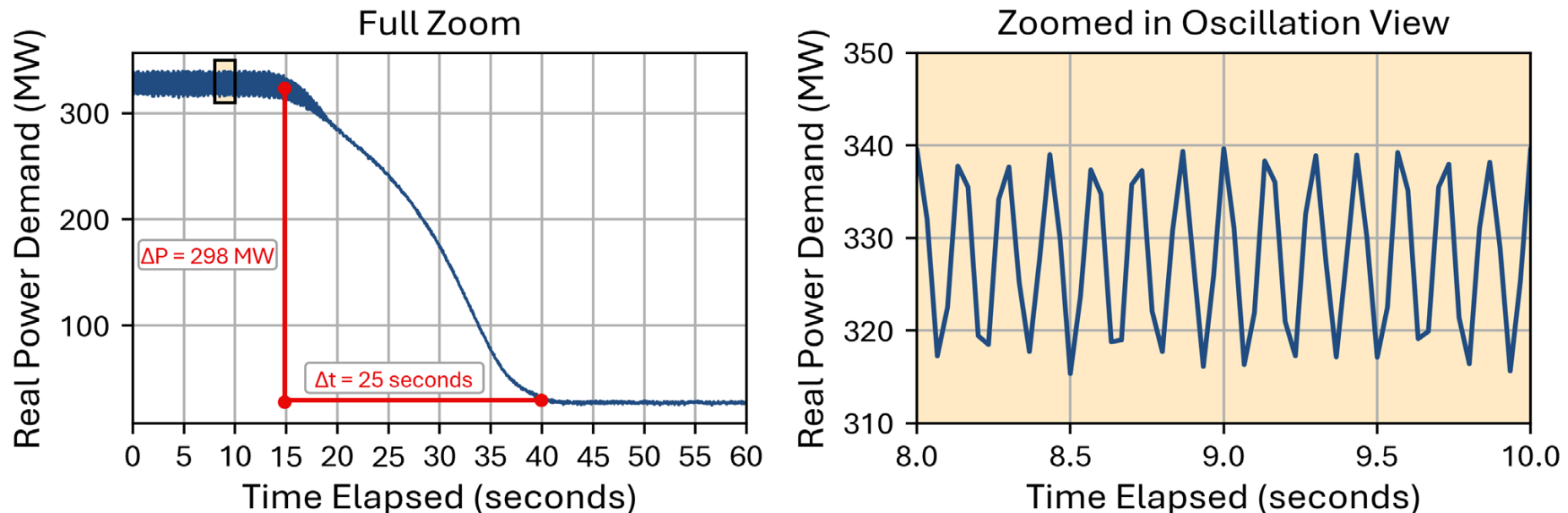
Additionally, the protection scheme needs to account for transmission breaker operations.

- Circuit breakers need 40 – 70ms to operate. and if the event detection timer in the Large Load protection scheme does not account for a breaker timing tolerance, then it may misoperate.
- For example, if the scheme is set to count a voltage depression at 80ms with a ± 20 ms tolerance, then it would operate if the voltage depression lasts for 60ms, which is within the time frame of the breaker operation.
- Therefore, the protection schemes should be updated to at least 80 – 120ms to account for breaker operation timing to reduce misoperation rates

Oscillation Risks

Data centers introduce new potential sources of oscillations on the power system. This is driven by both their repeated load fluctuations, which can cause forced oscillations, and their power electronic interface, which can cause resonance and converter-driven instability. Both can lead to uncontrolled generator tripping.

Oscillations have been seen in ERCOT and North America. Some of these have been sub-synchronous oscillations and could only be detected with high resolution measurements.



Data Center Modeling

The current generation of data center models have been shown to not accurately represent LCL behavior. Work is needed to create both positive sequence phasor domain models and electromagnetic transient models that properly represent LCL behavior. Any data center model must represent:

- the power electronic interfaces
- load profiles and resulting oscillations
- behavior, expected response, and controls of the data center

However, not all data centers are designed the same and this needs to be accounted for in the modeling.

- AI training, high-performance computing, and cryptocurrency mining will not always have UPSs installed for example

One of the challenges of creating accurate models is information availability.

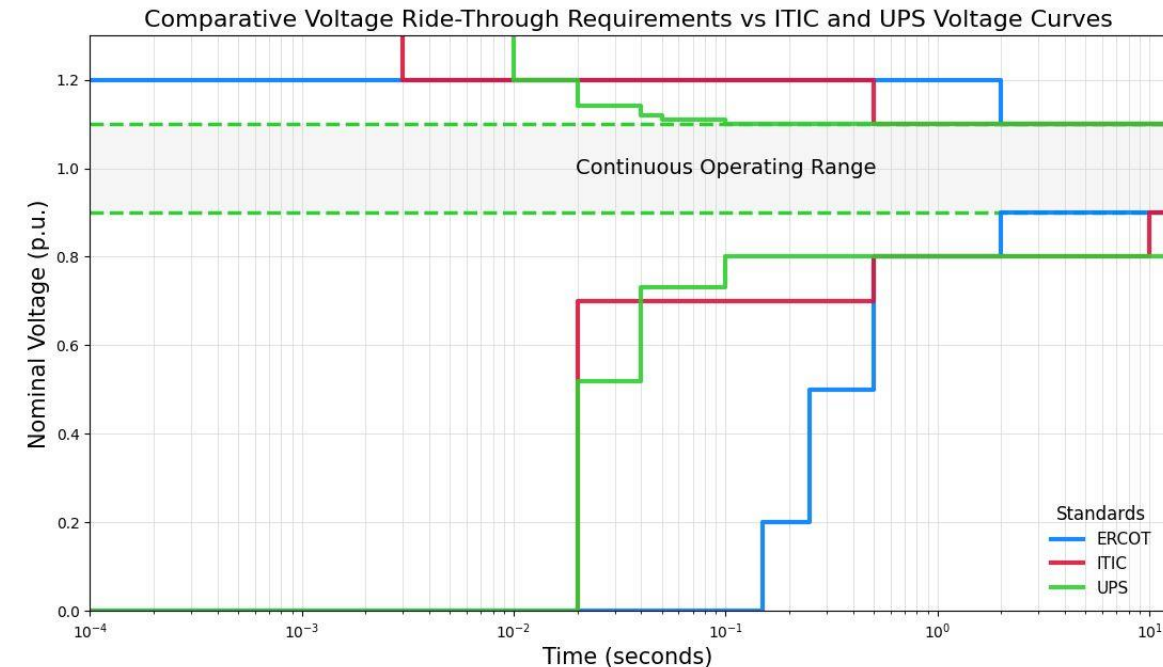
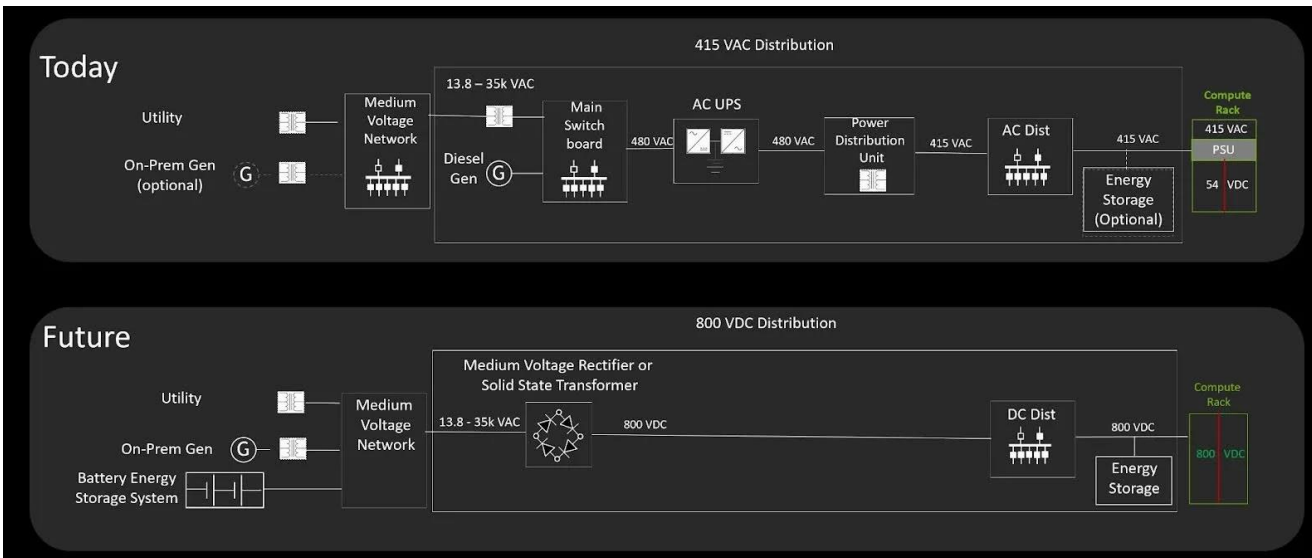
- Some of the information needs to come from equipment manufacturers as even the facility developer may not have it. However, the equipment manufacturers may not be familiar with what we need for power system studies.
- Some settings are software based and can be changed on the fly like with inverter-based resources. We need a process for capturing these changes and updating our models with them.

Finally, once the models are created, they should be verified against the events we've seen in the field.

The Evolving Data Center

As we grapple with ensuring grid reliability with the data centers currently deployed, their designs continue to evolve. New designs are pushing power consumption to 1 MW per rack while AI workloads are changing load profiles. To supply this power, new data center designs utilizing low voltage direct current (LVDC) are emerging - NVIDIA's 800VDC architecture and the OCP's Mt. Diablo $\pm 400\text{VDC}$ specification.

Rearchitecting data center power distribution systems will allow for more control over data center load while allowing for the innovations needed to close the performance gap between the current ITIC/UPS requirements versus emerging grid requirements.



Solid State Transformers

The next generation of data centers will likely use solid state transformers (SSTs) as the interface between the grid and the data center. SSTs use a power electronic interface to transform AC voltages, or convert AC to DC, provide active regulation of voltage and current.

When coupled with energy storage, SSTs can be used to meet emerging requirements such as ride-through and oscillation management by the smoothing of load fluctuations. They can also be capable of grid forming behaviors.

However, we have had numerous challenges with power electronic generation such as IBRs that we don't want with power electronic transformers. We learned that we need proper dynamic and EMT models for these generators and will need the same for these SSTs. We also need to see what their protection and fault behavior is and study how that interacts with the grid.

Finally, we need to ensure that grid codes are updated as the transformer moves from a passive device to an actively controlled power electronic device.



Source: <https://www.dgmatrix.com/>



Source: <https://heronpower.com/product/data-centers>

Implications for the Grid

These new data center designs will require us to continue to update models, and coordinate with LCLs as these facilities begin to become more controllable. The current discourse is around modeling down to the UPS system but with facility level power transformation with SSTs, hopefully we can abstract the modeling detail to a higher level. Larger facilities especially when co-located with generation begin to resemble a large scale microgrid. Coordination with the grid become paramount.

- Controls and protection behavior must be coordinated.
- Grid forming behavior must be studied and tuned.
- Grid codes must be updated.

Once performance requirements are determined we need data centers to meet these requirements.

- However, the design and manufacturing timeline for equipment is 2+ years. For in-service facilities and those being built before requirements were implemented retrofits will be needed to comply with grid requirements.
- For new designs, requirements can be incorporated but these new designs will not be deployed until 2028 for most equipment.

Key Takeaway: The choice of technologies selected to address the identified reliability issues can also cause new issues as well. It will be critical to monitor evolving data center designs to ensure that new reliability issues do not emerge. Given the pace of LCL development, and design and manufacturing timelines, time is not on our side. We must move quickly.

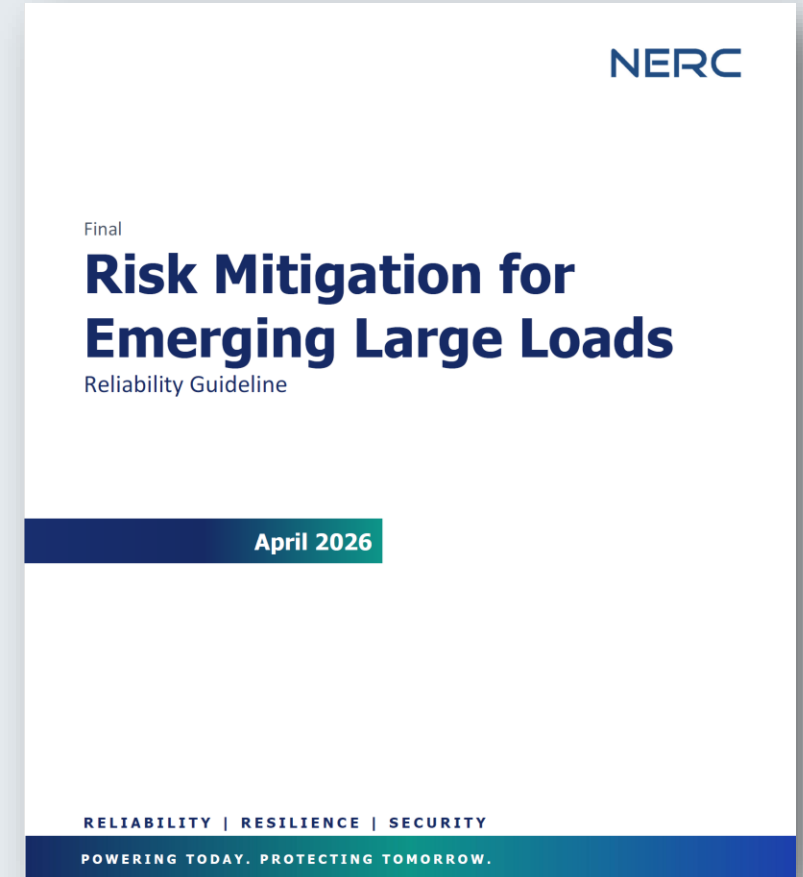
Reliability Guideline – Risk Mitigation for Emerging Large Loads

The Reliability Guideline provides best-practice guidance on mitigating risks to BPS reliability from integration of emerging large loads.

Recommendations for all registered entities and large load entities covers:

- Data collection and modeling
- Interconnection and planning study processes
- Commissioning processes & operational considerations
- Security and resilience
- High-speed monitoring

Key Takeaway: The Reliability Guideline provides best practices that should be adopted across the industry to mitigate the risks posed by emerging large loads. Entities should review it and implement its recommendations in their organizations to safeguard BPS reliability.



Source: https://www.nerc.com/globalassets/our-work/guidelines/reliability/RG_Risk-Mitigation-For-Emerging-Large-Loads.pdf

We Can't Do It Alone

LCLs are power electronic-based, which have different characteristics when compared to historical Large Loads that were more motor-driven.

Most utilities and system operators are not experts in power electronics. Therefore, to fully understand these LCLs and the risks they pose to the grid, grid operators need to collaborate with LCL entities and equipment manufacturers and draw upon expertise within national labs and universities. Engagement with groups like the Open Compute Project is critical so grid requirements are embedded in all levels of LCL design.

New standards such as IEEE 4200 and NERCs Project 2026-02 CLE-001, CLE-002, and CLE-003 are key.

NERC Large Load
Working Group

Energy Systems
Integration Group Large
Load Task Force

IEEE Data Centers:
Standards Needs
Analysis and
Recommendations

NERC Load Modeling
Working Group

ERCOT Large Load
Working Group

Other IEEE and CIGRE
task forces and working
groups

Questions/Comments?

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